

FIG. C-155 Labyrinth shaft seals. (Source: Sulzer-Burckhardt.)

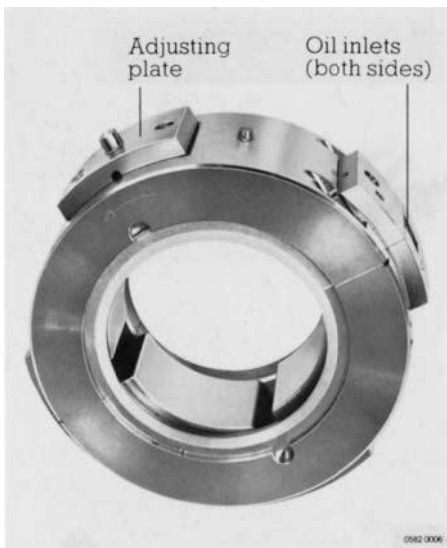


FIG. C-156 Multisegment journal bearing with four tilting pads. (Source: Sulzer-Burckhardt.)

due to unbalance, the friction-induced heat is immediately passed to the massive stator, thus avoiding distortion of the rotor. Gas-tight shaft seals and standstill seals can be fitted for special requirements. See Fig. C-155.

Journal and axial bearings

Journal bearings. In the standard version, i.e., with the compressor rotor solidly coupled and the rotor thrust transferred to the axial thrust bearing of the prime mover or the gear, the bearing housings are equipped only with journal bearings. Two-lobe bearings are provided for the lower speed range; tilting pad journal bearings are generally used for the higher speeds of the smaller frame sizes for reasons of stability. The slight curvature of the adjusting plates allows the bearings to be set accurately on erection. The bearings are firmly held in position by the bearing housing top half. See Figs. C-156 through C-158.

Two-lobe bearings are suitable for both senses of rotation, while tilting pad bearings are essentially for only one direction, although they can tolerate running backward with a somewhat reduced load capacity.

Axial thrust bearings. If a flexible coupling is selected between driver and driven machine, the bearing housing can accommodate the necessary tilting pad thrust bearing. The purpose of this bearing is to absorb the remaining thrust of the machine and any significant axial friction thrust of the coupling due to sharp

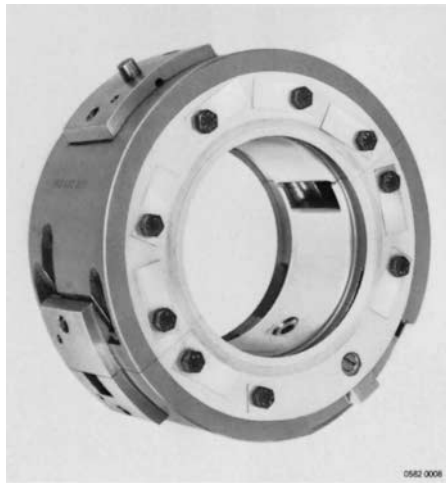


FIG. C-157 Two-lobe journal bearing. (Source: Sulzer-Burckhardt.)

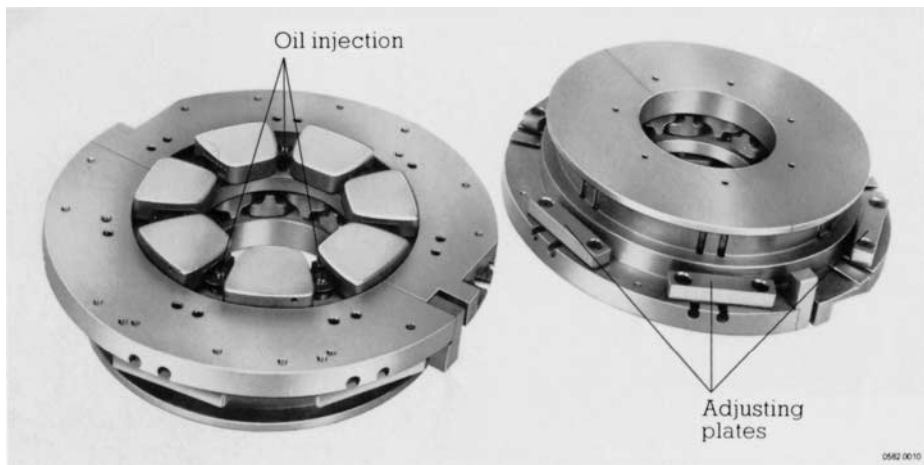


FIG. C-158 Kingsbury-type thrust bearing with self-equalizing pads with directed lubrication. (Source: Sulzer-Burckhardt.)

temporary differential expansion between rotor and casing. To provide easy access and reduce the overhang, it is preferably mounted on the free shaft end.

The tilting pads are supported on load-equalizing segments that allow angularity of the shaft up to 0.3 percent.

Because the tilting pads are supported eccentrically, thrust bearings are suitable for only one direction but tolerate a reversed rotation at a reduced load capacity.

Solid coupling. It is this OEM's normal practice to make extensive use of solid couplings allowing the use of only one axial thrust bearing for single- or multiple-casing arrangements.

An intermediate shaft, flexible enough to allow for considerable misalignment, is inserted between the two shaft ends of the machines to be coupled together (Fig. C-159). In case of motor-driven units, the normal technique is to use single helical gears provided with thrust collars on the pinion shaft, as shown in Figs. C-160 and C-161. The thrust collars not only neutralize the axial thrust created by the meshing of the teeth cut at an angle to the axis of the shaft, but also transmit

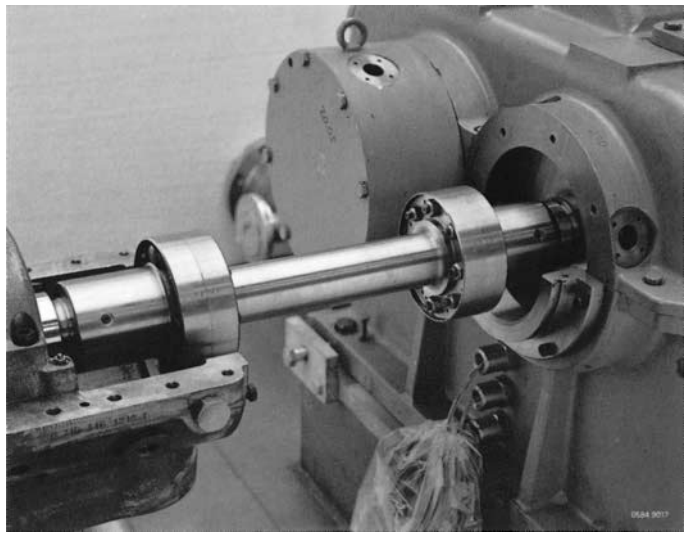


FIG. C-159 The solid quill-shaft coupling conforms to API 671 standard and consists of the quill shaft and the two hubs hydraulically fitted onto the shaft ends of the connected machines. On each coupling side, an equal number of tie bolts for axial fixation and tapered dowel pins for torque transmission and centering ensure a clearly defined connection. Balancing as a complete assembled unit and correlative marking enable removal and remounting of this intermediate shaft with the connected rotors remaining in place, without affecting the balancing quality and vibration behavior of the complete string. (Source: Sulzer-Burckhardt.)

the unbalanced axial thrust of the high-speed rotor train to the thrust bearing on the low-speed wheel shaft. Good gear meshing requires parallelity of gear and pinion shaft and automatically ensures parallelity of the contact surfaces of thrust collar and wheel rim. The slight tapering of the thrust collars is responsible for the formation of a wedge-type oil film creating a pressure zone spread out on an enlarged surface with a pressure distribution very similar to that of a standard oil-lubricated journal bearing.

The relative motion between the two contact surfaces of the thrust collar system is a combination of rolling and sliding and takes place near the pitch circle diameter, resulting in a very small relative velocity. The thrust transmission is therefore conducted with minimal mechanical losses. The considerably reduced losses of the single thrust bearing on the low-speed shaft as compared with the high losses of individual thrust bearings on the high-speed train lead to a substantial power saving. This low-speed bearing can also be more amply dimensioned to provide a much higher overload capacity.

For direct turbine-driven compressor trains, the thrust bearing is usually located in the turbine. Also in this case solid couplings with flexible intermediate shafts are much preferred.

This coupling arrangment avoids heavy overhung gear couplings that are usually responsible for not clearly defined lower critical speeds and for the phenomena of torque lock leading to additional loading of the axial thrust bearing. The resulting axial friction forces can become quite substantial if insufficient attention is given to the cleanliness of the lubricating oil.

Centrifugal compressors

Centrifugal compressors made by this information source are available for the compression of various gases within a flow range of about 0.6 to 60 m^3/s and pressures up to 700 bar. They are used in the most varied applications, mainly in

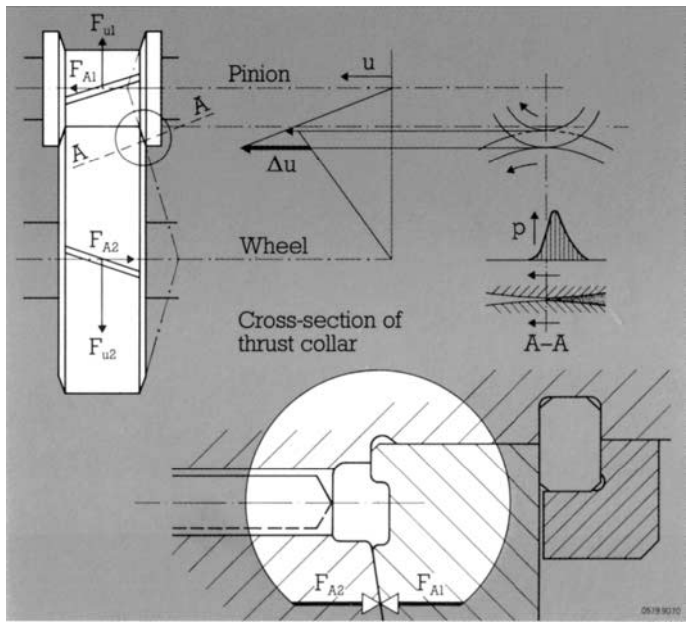


FIG. C-160 Method of axial thrust transfer in a single helical gear with thrust collar. (Source: Sulzer-Burckhardt.)

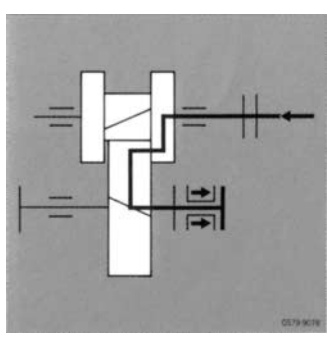


FIG. C-161 Transfer of external forces. (Source: Sulzer-Burckhardt.)

the fields of process engineering, chemical industry, and mineral ore processing, as well as oil and gas production. See Fig. C-127.

Units made by this information source have power inputs varying between a few hundred kilowatts and nearly 40 MW.

To minimize design lead time and ensure economical manufacturing and stocking of the major components, the range of these compressors has been standardized to an optimum extent. The constant geometrical ratio between the frame sizes allows compressors to be assembled to different specifications with the minimum inventory of component parts, diminishing the investment on spare parts. It also permits comparison and forecast of performance and stresses within the series of frame sizes. Figure C-162 shows a section through a centrifugal axially split compressor.

Type range. The standardized range of horizontally split casing types (see Fig. C-131) extends to eight frame sizes for this OEM source including three pressure ranges for each frame size up to 70 bar and is designed for appropriate turndown.

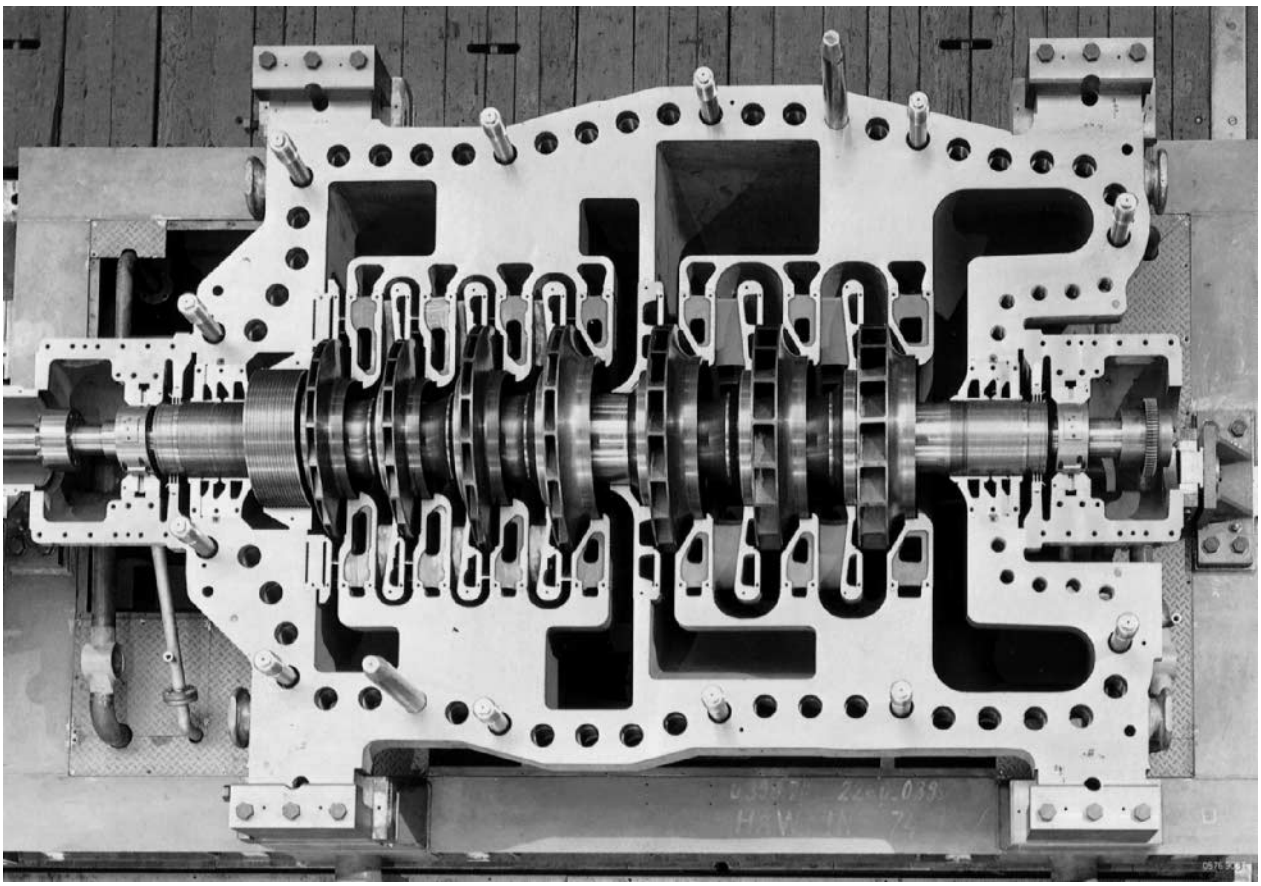


FIG. C-162 Section through a horizontally split axial compressor. (Source: Sulzer-Burckhardt.)

An appropriate selection of impellers and diffusers enables the compressor stages to be matched to any specified operating data. See Figs. C-163 through C-167.

The barrel-type design is suitable for discharge pressures of up to 700 bar. It has vertically split casing with end cover and autoclave cover and horizontally split internal casing for easy assembly and dismantling. Option of intermediate nozzles for connecting to intercoolers or for side-stream intake or extraction. The internal components, including the horizontally split internal casing, autoclave cover, diaphragms, and rotor with bearings and seals, are assembled outside the barrel casing; the internal clearances can then be checked exactly prior to final assembly.

Compressor design: adaptability of standardized product range. The different ways of adapting standard designs to varied operating conditions are described as follows:

Adapting to different specific flow conditions. The basic dimensions of the components and parts of compressors such as casings, impellers, diffusers, and bearings are standardized by using a constant scale factor between the different frame sizes. As can be seen in Fig. C-165, this factor is 1.25 for the casing sizes, 1.12 for impeller diameters, and even smaller for certain dimensions of internal parts. This system permits the assembly of various compressors complying to specified data by using a minimum number of predesigned parts. Moreover, it ensures accurate forecasting of performance interpolated from other frame sizes. The eight frame sizes cover a range between 0.5 and 60 m^3/s. The range of higher

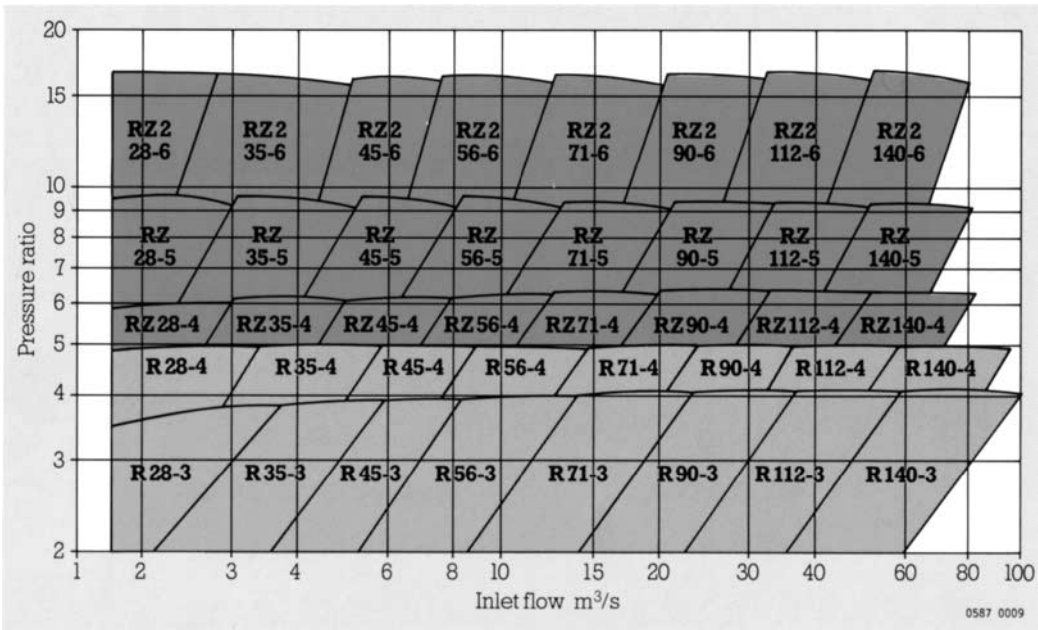


FIG. C-163 Selection chart for centrifugal compressors of the R and RZ series. (Source: Sulzer-Burckhardt.)

suction volumes above about 25 m^3/s and up to 350 m^3/s is covered by 12 frame sizes of axial compressors, where a scale factor of 1.12 is used.

Adapting to different specified pressures. This information source's models (R, RZ, and RS) can be equipped with casings made of gray cast iron, nodular cast iron, or cast steel which, depending on the frame size, makes them suitable for standard design pressure classes of 6, 16, 25, 40, or 64 bar. The RB, RBZ, and RBS casings are invariably made of fabricated or cast steel to cover the standard pressure range of up to 700 bar.

Adapting to different process conditions. The process industry has an ever-changing range of requirements concerning the arrangement of external casing nozzles, either when intercooling is needed to limit temperature during compression or intermediate inlets have to be provided for side-streams. The standard design is very flexible in this respect.

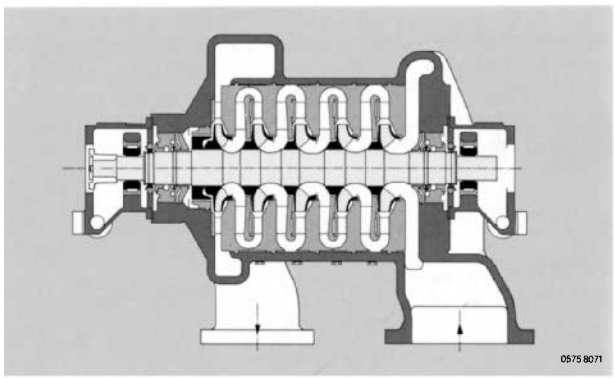
CENTRIFUGALS WITH HORIZONTALLY SPLIT CASINGS

- Series R: straight-through compressor without provision for intercooling
- Series RZ, RZ2: compressors with one or two pairs of intermediate nozzles for connecting to one or two intercoolers
- Series RS, RS2: compressors with one or two intermediate inlet nozzles for one or two side-streams

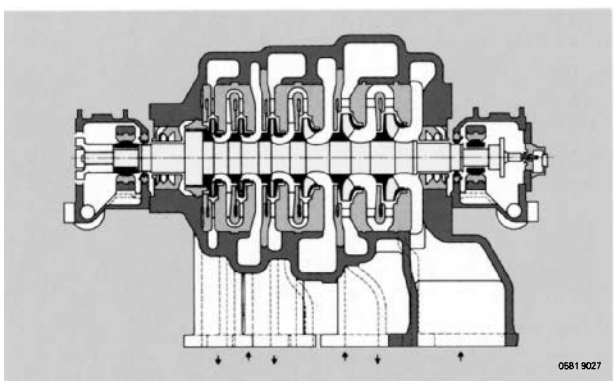
CENTRIFUGALS WITH VERTICALLY SPLIT CASINGS

- Series RB: straight-through compressor without provision for intercooling
- Series RBZ: compressor with one pair of intermediate nozzles for connection to an intercooler
- Series RBS: compressor RB with additional inlet or outlet

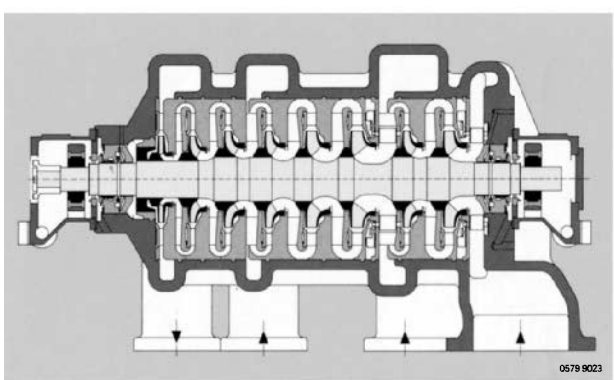
Series R, RZ, RS



Series R:
machines of one to eight stages with horizontally split casing

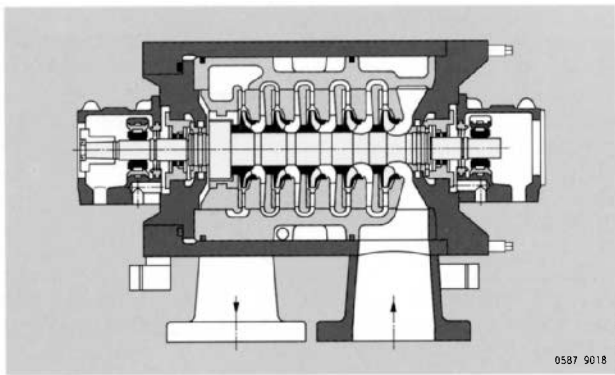


Series RZ:
machines of four to eight stages with horizontally split casing, one to two pairs of intermediate nozzles generally for connecting external intercoolers

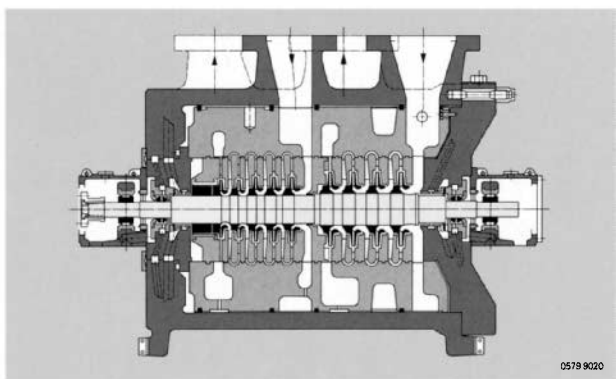


Series RS:
machines of four to eight stages with horizontally split casing and one or two nozzles for side stream intake or extraction

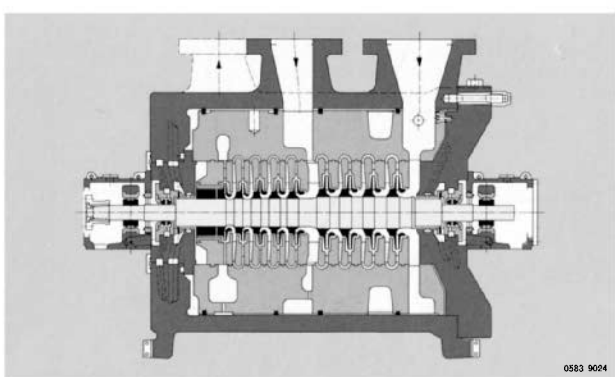
Series RB, RBZ, RBS



Series RB:
machines of two to eight stages with vertically split casing (barrel type)



Series RBZ:
machines of four to nine stages with vertically split casing and one pair of intermediate nozzles



Series RBS:
machines of four to nine stages with vertically split casing and additional inlet or outlet nozzles

FIG. C-164 **Typical series designations for centrifugal compressors. (Source: Sulzer-Burckhardt.)**

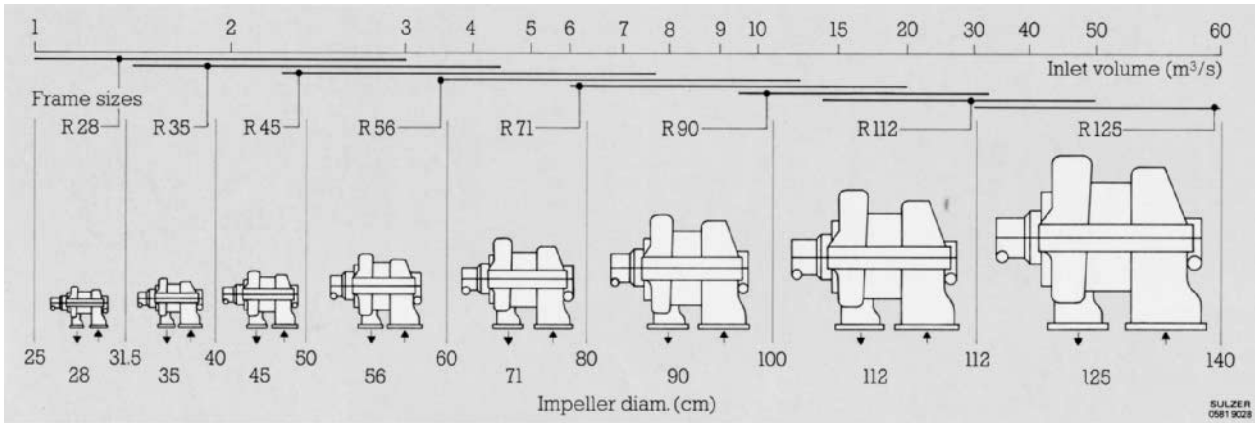


FIG. C-165 Standardized range of centrifugal compressors based on eight geometrically similar frame sizes. Each frame is designed to accommodate three different impeller diameters and different impeller types. (Source: Sulzer-Burckhardt.)

	Compressor types															
	R, RZ, RS								RB, RBZ, RBS							
Frame sizes	28	35	45	56	71	90	112	125	28	31	35	40	45	56	71	90
Pressure classes																
6					▭	▭	▭	▭								
16	▭	▭	▭	▭	▭	▭	▭	▭								
25					▭	▭	▭	▭								▭
40	▭	▭	▭	▭					▭	▭	▭	▭	▭	▭	▭	▭
64	▭	▭			▭				▭	▭	▭	▭	▭	▭	▭	▭
100									▭	▭	▭	▭	▭	▭	▭	▭
160									▭	▭	▭	▭	▭	▭		
250									▭	▭	▭	▭	▭	▭		
320									▭	▭	▭	▭	▭	▭		
400									▭	▭	▭	▭	▭			
500									▭	▭	▭					
700									▭	▭	▭					

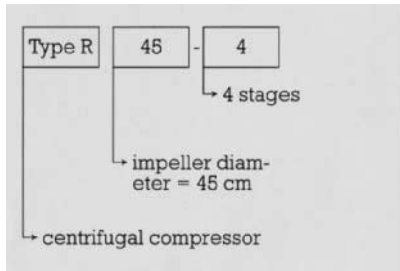
FIG. C-166 Standard pressure classes. (Source: Sulzer-Burckhardt.)

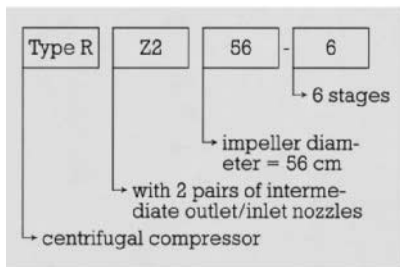
Building block system. Each compressor frame size exists in different standardized lengths to accommodate different numbers of stages. With cast casings, adjustments of the pattern are made using a modular technique whereby spacing rings are fitted between standardized pattern parts of the casing to alter the length (Fig. C-167).

When the head required is more than that practicable within one compressor body, two or more casings can be connected in series to form a train.

Design features

Casing. Depending on the required pressure class and the type of gas, the horizontally split casings of the series R, RZ, and RS are made of gray cast iron, nodular cast iron, or cast steel, unalloyed or alloyed. See Fig. C-168 for a section through a horizontally split centrifugal compressor and Fig. C-169 for a section through a barrel centrifugal compressor. All suction, intermediate, and discharge nozzles are normally facing downward to facilitate inspection without disturbing the process pipe connections. Optionally they can be arranged facing upward. In the case of flammable or toxic gases, the horizontal division flange can be provided





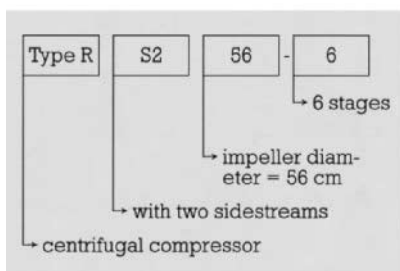


FIG. C-167A This is an illustration of the building block technique for adapting standard casing parts of a given frame size to various process conditions. For the casings of the three series R, RZ, and RS the same basic patterns are used. All casings are made available with nozzles facing upward. (Source: Sulzer-Burckhardt.)

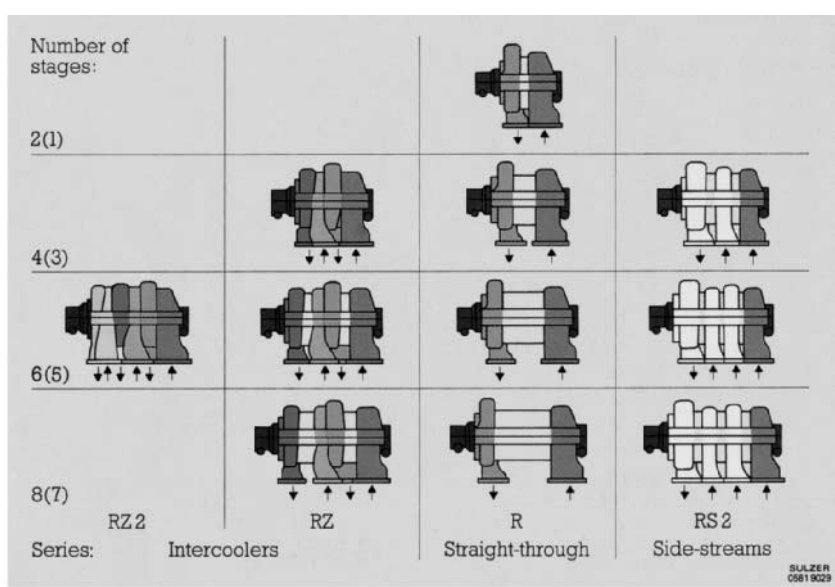


FIG. C-167B For barrel-type compressors the same designations apply as in Fig. 167A although "R" is replaced by "RB." (Source: Sulzer-Burckhardt.)

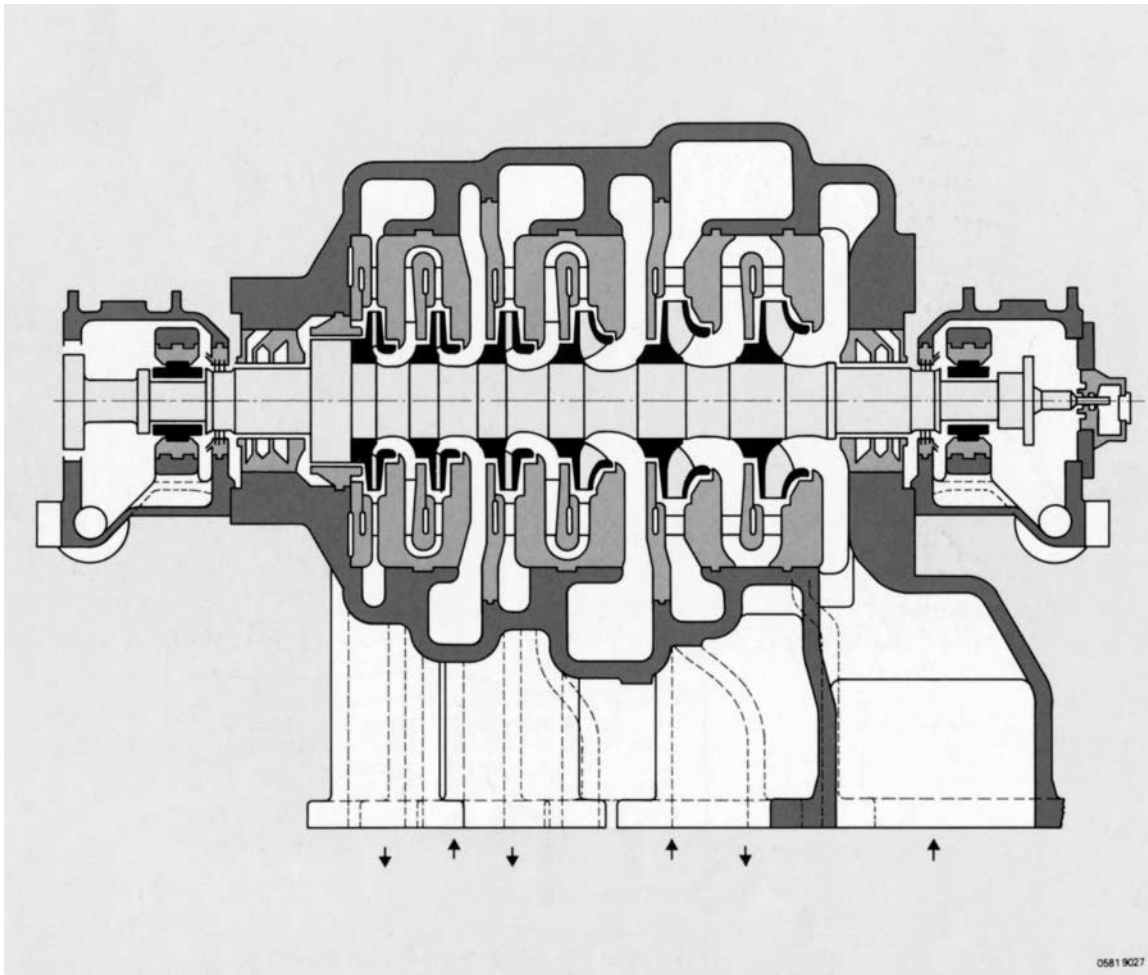


FIG. C-168 Cross-section of horizontally split compressor. (Source: Sulzer-Burckhardt.)

with drainage grooves to allow controlled leakage or inert gas sealing. Vertically split barrel casings of the series RB, RBZ, and RBS are made of cast steel or fabricated steel with the nozzles welded on. The barrel is closed at one end by an internal autoclave-type cover tightly locked by the inner gas pressure toward the shoulder of the outer casing. At the other end the cover is bolted to the casing and sealed by O-rings.

The diaphragms of both the R and RB series are normally made of gray or nodular cast iron, the diffusers of steel with the vanes welded on. The casing is supported by four feet at the horizontal flange to avoid misalignment due to thermal expansion.

The internal parts of the barrel compressor consist of the same standard components as are used for the R series. The horizontally split inner casing with rotor, diaphragms, diffusers, autoclave, end cover, and bolted-on bearing housings is preassembled outside the pressure casing. This subassembly is then inserted into the barrel by means of guide rails. See Figs. C-170 and C-171.

See Fig. C-172 for diagrammatic representation of impellers on rotor assembly, as well as illustration of specific impeller design features. The numbers in the figure correspond with the design features listed in the table on p. C-175.

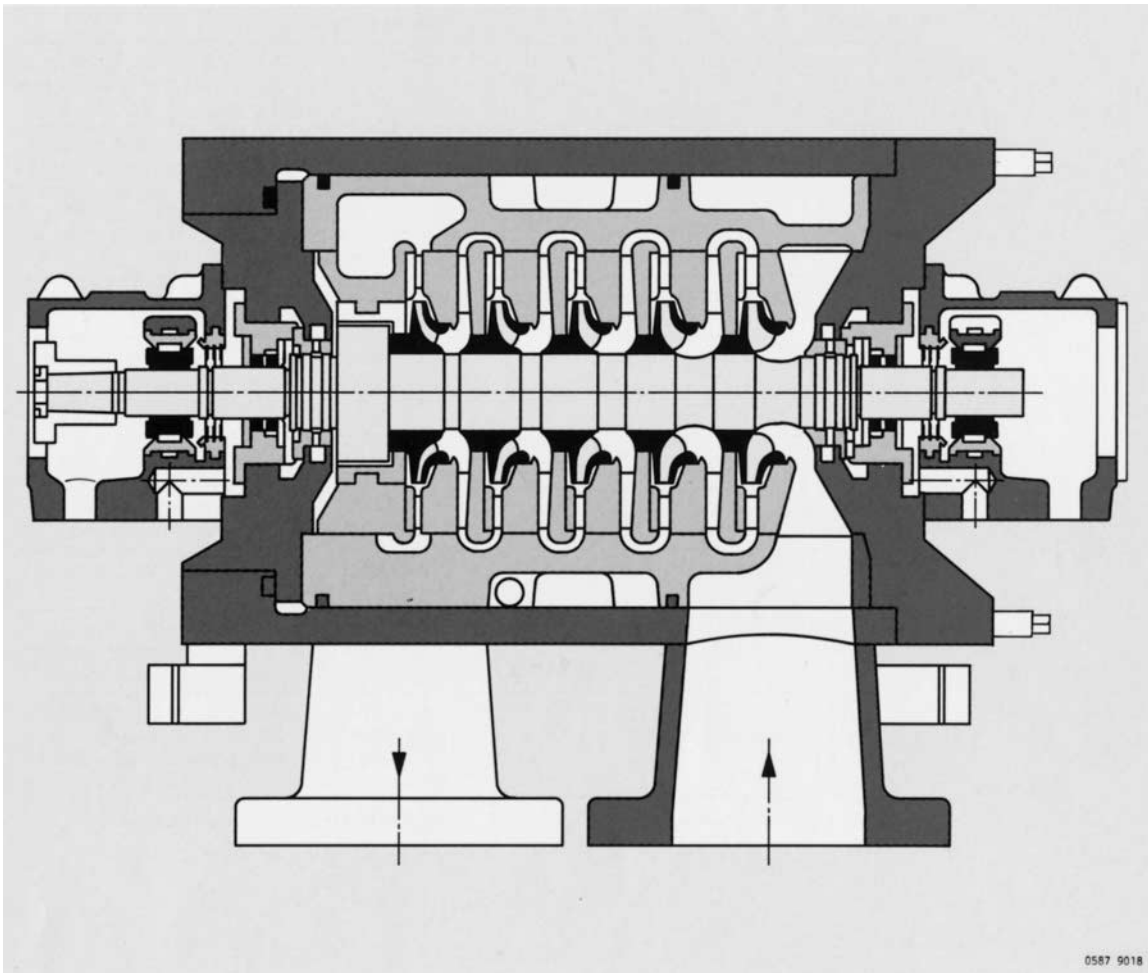


FIG. C-169 Cross-section of barrel compressor. (Source: Sulzer-Burckhardt.)

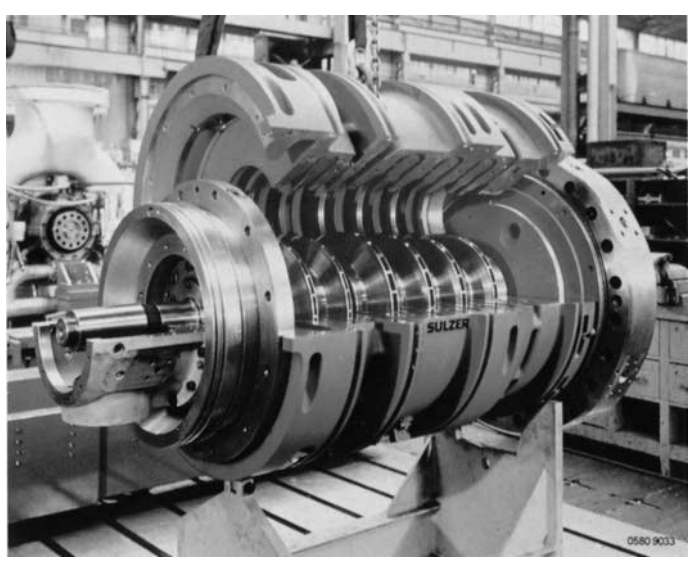


FIG. C-170 Inner subassembly including end covers, seals, and bearing housings. (Source: Sulzer-Burckhardt.)

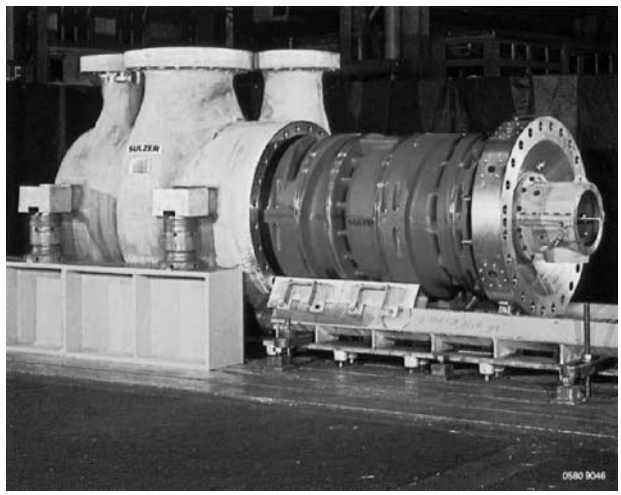


FIG. C-171 The inner subassembly is inserted into the barrel casing by means of guide rails. (Source: Sulzer-Burckhardt.)

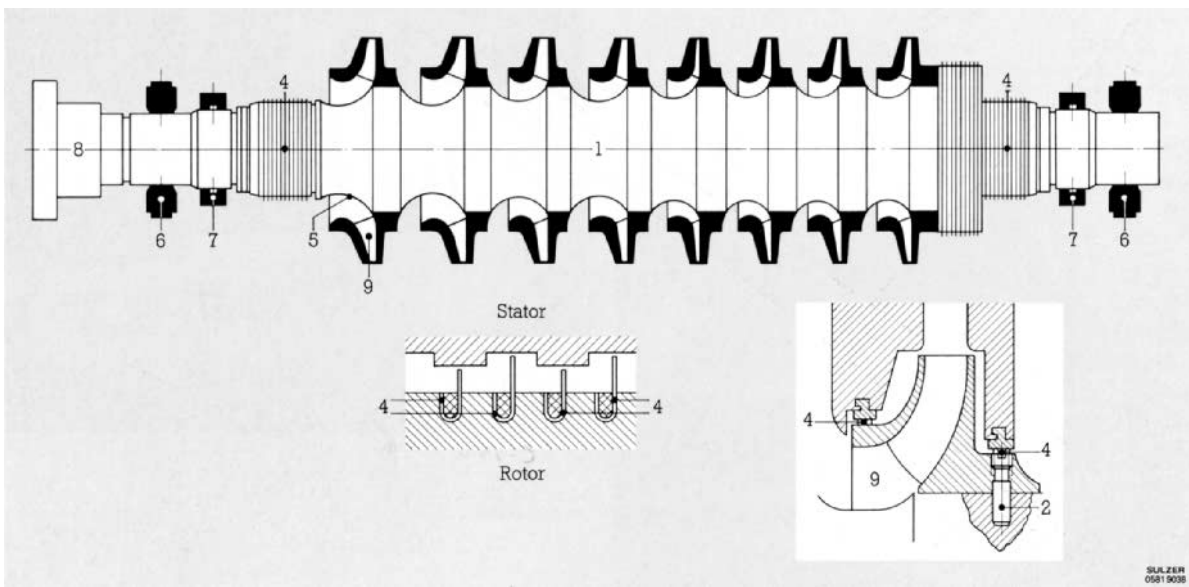


FIG. C-172 Design principles of turbocompressor rotors. (Source: Sulzer-Burckhardt.)

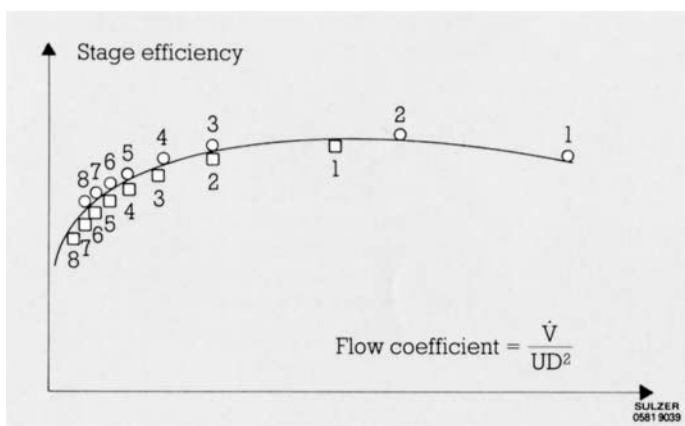


FIG. C-173 Influence of flow coefficient of the first impeller on the efficiency of subsequent stages. ○, efficiency distribution of a compressor with a high-flow impeller at suction. □, efficiency distribution of a compressor with a medium-flow impeller at suction. (Source: Sulzer-Burckhardt.)

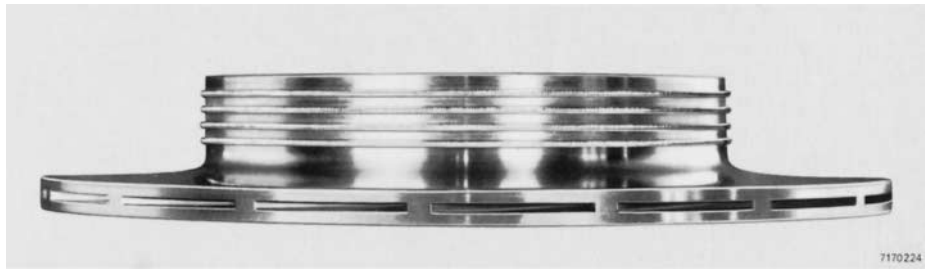


FIG. C-174 Very narrow impeller. Blades milled out of the hub disc and brazed to cover disc. (Source: Sulzer-Burckhardt.)

Design Features	Aimed at Providing
1. Solid sturdy rotor; integral dummy piston for high-speed and high-pressure applications	Minimum sensitivity to critical speeds and unbalance due to higher rotor stability; reduction of rotor internal damping
2. Shrink fit secured by symmetrically arranged radial dowels for impellers	No need for keys and distance bushings; fixation ensures concentricity and perfect balance under extreme operating conditions; allows larger shaft diameters; reduces stress on shaft and impeller
3. No shaft sleeves between stages	Reduces rotor hysteresis and increases running stability
4. Labyrinths always on the rotating element	No distortion of rotor due to local heating up in case of rubbing; labyrinths can be refitted easily
5. Nickel or other plating of shaft portions exposed to corrosion or entirely stainless shafts with mild steel welding or plating at bearings and at floating ring seals	Plating instead of shaft sleeves is a more direct protection; allows larger shaft diameters
6. Tilting pad radial bearings for higher speeds and pressures	Improves running stability; no oil-whip; higher external damping
7. No shaft sleeves for liquid-film seals	Minimum wear; perfect concentricity of shaft and rings with resultant minimum clearances and seal medium losses
8. Solid coupling, tightly bolted to flexible intermediate shaft	Improves reliability due to elimination of high-speed thrust bearing and toothed-type couplings; no gear lock thrust on high-speed thrust bearing
9. High-flow impeller at suction	Improves overall efficiency

Impellers. Impellers with high flow coefficients allow smaller diameters and optimum performance in all stages, resulting in high overall efficiencies (see Fig. C-173). These impellers are of fully welded construction with the blades shaped in three dimensions. A continuous welding technique ensures good aerodynamics and uniform stress distribution.



FIG. C-175 Wide impeller. Blades welded to both hub disc and cover disc. (Source: Sulzer-Burckhardt.)

Very narrow impellers consist of a hub disc with integral blades milled out of the disc material and a hub disc brazed to the blades according to a special manufacturing procedure (see Fig. C-174). For wider impellers, still not allowing welding on hub and cover disc, the blades are welded only to the cover disc (see Fig. C-175), and thereafter the hub disc is brazed to the blades (see Fig. C-176). Both alternatives provide accurate flow passages and a very smooth surface. In those rare cases where brazing is considered incompatible with the nature of the gas, such narrow impellers can be riveted.

Solid coupling. It is normal practice for some manufacturers, such as this information source, to make extensive use of solid couplings allowing the use of only one axial thrust bearing for single- or multiple-casing arrangements.

An intermediate shaft, flexible enough to allow for considerable misalignment, is inserted between the two shaft ends of the machines to be coupled together (Fig. C-177). In case of motor-driven units, the normal technique is to use single helical gears provided with thrust collars on the pinion shaft, as shown in Figs. C-178 and C-179. The thrust collars not only neutralize the axial thrust created by the meshing of the teeth cut at an angle to the axis of the shaft, but also transmit the unbalanced axial thrust of the high-speed rotor train to the thrust bearing on the low-speed wheel shaft.

Good gear meshing requires parallelity of gear and pinion shaft and automatically ensures parallelity of the contact surfaces of thrust collar and wheel rim. The slight tapering of the thrust collars is responsible for the formation of a wedge-type oil film creating a pressure zone spread out on an enlarged surface with a pressure distribution very similar to that of a standard-oil-lubricated journal bearing.

The relative motion between the two contact surfaces of the thrust collar system is a combination of rolling and sliding and takes place near the pitch circle diameter,

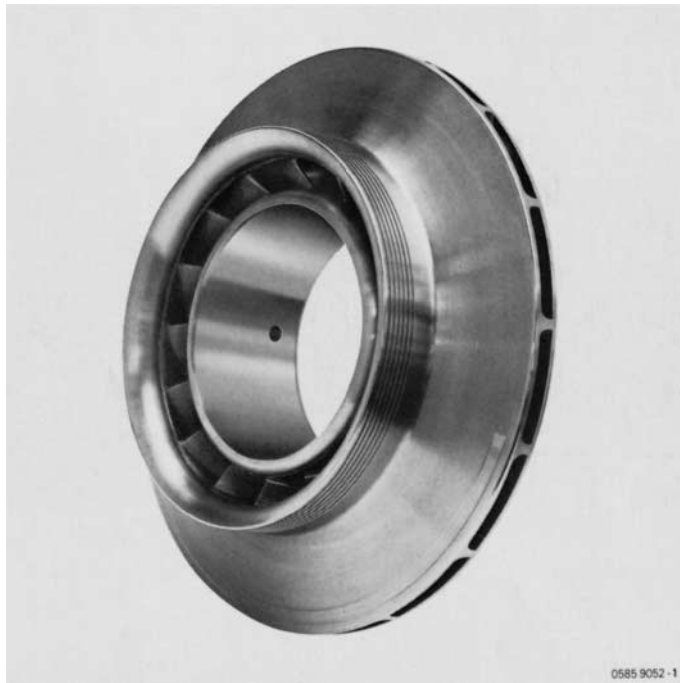


FIG. C-176 Narrow impeller. Blades welded to cover disc and brazed to hub disc. (Source: Sulzer-Burckhardt.)

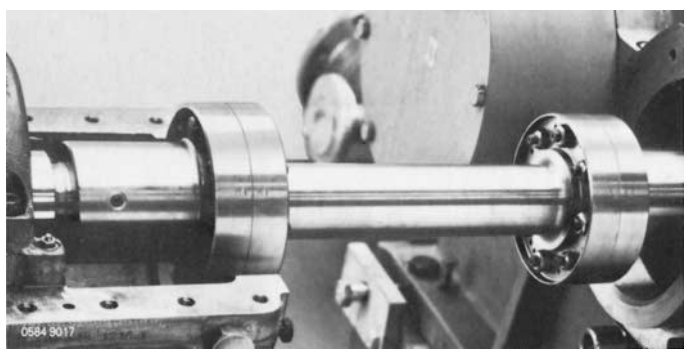


FIG. C-177 The solid quill-shaft coupling conforms to API 671 standard and consists of the quill shaft and the two hubs hydraulically fitted onto the shaft ends of the connected machines. On each coupling side, an equal number of tie bolts for axial fixation and tapered dowel pins for torque transmission and centering ensure a clearly defined connection. Balancing as a complete assembled unit and correlative marking enable removal and remounting of this intermediate shaft with the connected rotors remaining in place, without affecting the balancing quality and vibration behavior of the complete string. (Source: Sulzer-Burckhardt.)

resulting in a small relative velocity. The thrust transmission is therefore effected with almost no mechanical losses. The considerably reduced losses of the single thrust bearing on the low-speed shaft as compared with the high losses of individual thrust bearings on the high-speed train lead to a substantial power saving. Moreover, this low-speed bearing can be more amply dimensioned to provide a much higher overload capacity.

For direct turbine-driven compressor trains, the thrust bearing is usually located in the turbine. Also in this case solid couplings with flexible intermediate shafts are much preferred.

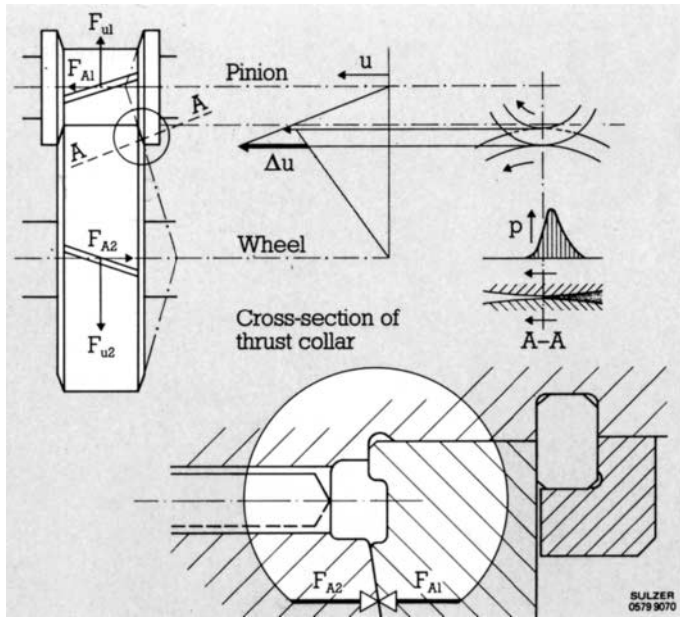


FIG. C-178 Method of axial thrust transfer in a single helical gear with thrust collar. F_u = peripheral force, F_A = axial force, u = peripheral speed, p = pressure. (Source: Sulzer-Burckhardt.)

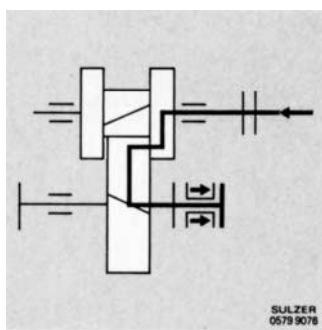


FIG. C-179 Transfer of external forces in bearing. (Source: Sulzer-Burckhardt.)

This coupling arrangement avoids heavy overhung gear couplings that are usually responsible for not clearly defined lower critical speeds and for the phenomena of torque lock leading to additional loading of the axial thrust bearing. As with axial compressors, the resulting axial friction forces can become quite substantial if insufficient attention is given to the cleanliness of the lubricating oil.

Journal and axial bearings

Journal bearings. See Fig. C-180. In the normal version, i.e., with the compressor rotor solidly coupled and the rotor thrust transferred to the axial thrust bearing of the prime mover or the gear, the bearing housings are equipped only with journal bearings. Two-lobe bearings are provided for speeds up to about 7000 rpm, tilting pad journal bearings are generally used for higher speeds and for types RB, RBZ, and RBS for reasons of stability. The slight curvature of the adjusting plates allows the bearings to be set accurately on installation. The bearings are firmly held in position by the bearing housing top half.



FIG. C-180 Multisegment journal bearing with four tilting pads. (Source: Sulzer-Burckhardt.)

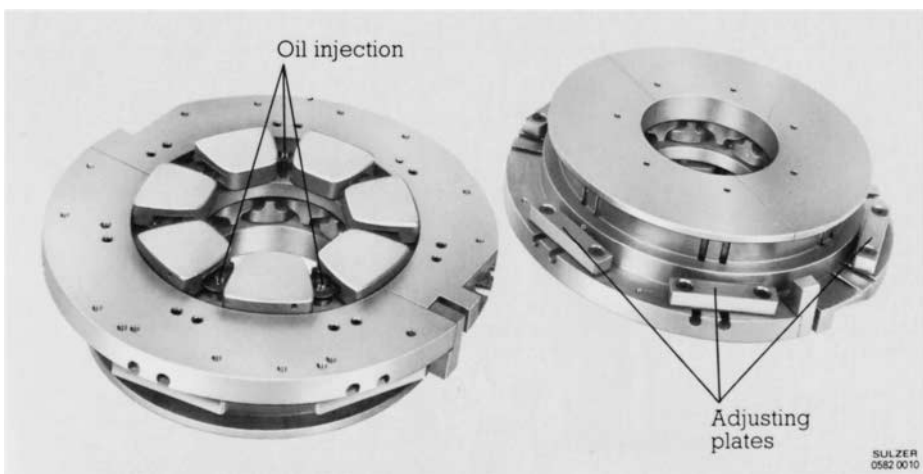


FIG. C-181 Kingsbury-type thrust bearing with self-equalizing pads with directed lubrication. (Source: Sulzer-Burckhardt.)

Two-lobe bearings are suitable for both senses of rotation, while tilting pad bearings are essentially for only one direction, although they can tolerate running backward with a somewhat reduced load capacity.

Axial thrust bearings. See Fig. C-181. Compressors driven through toothed or flexible couplings have a special bearing housing that can accommodate the necessary additional tilting pad thrust bearing. The purpose of this bearing is to absorb the remaining thrust of the machine and any significant axial friction thrust of the coupling due to sharp temporary differential expansion between rotor and casing. To provide easy access and reduce the overhang, it is preferable to mount it on the free shaft end.

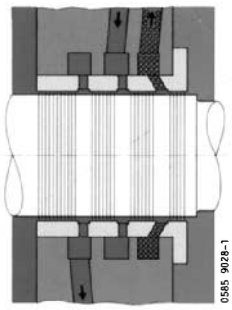
The tilting pads are supported on load-equalizing segments that allow angularity of the shaft up to 0.3 percent.

Because the tilting pads are supported eccentrically, thrust bearings are suitable for only one direction but tolerate a reversed rotation at a somewhat reduced load capacity.

Labyrinth seal

For low pressures or for applications, where either a constant loss of the process gas to atmosphere or a constant slight contamination of the process gas with buffer gas can be tolerated, or where even at high pressures the oilseal must be excluded for reasons of incompatibility of oil with the process gas, the appropriate seal is the labyrinth type. The simplicity of its design and associated system make it a reliable element with easy maintenance.

By interchanging the connection for buffer gas and ejector, it either prevents leakage to atmosphere (picture) or contamination of process gas.



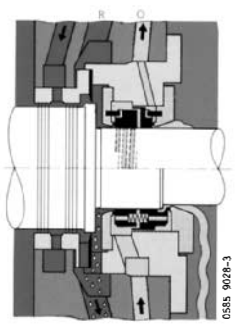
The labyrinths consist of thin stainless J-shaped steel strips on the rotating part and are kept in place by a caulked in steel wire. All strips are easily replaceable.

The labyrinth strips are caulked into the rotating part because, if rubbing occurs, the thin metal is then abraded without the casing and the rotor becoming unacceptably warm, avoiding rotor distortion.

Liquid film seal

Gas-tight shaft seals of the non-contact type are SULZER's basic recommendation for compressors delivering inflammable, toxic or costly gases.
Stationary floating rings operate satisfactorily at peripheral speeds as high as 100 m/s and are therefore preferably fitted on high-speed machines or compressors with large shaft diameters.

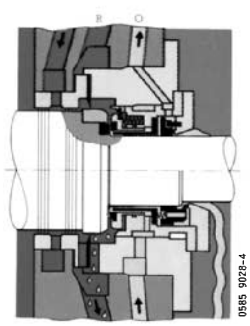
Liquid film seals consist of a pair of seal rings fitted in such a way that they are floating to follow radial movements of the shaft relative to the casing.
One ring of a pair is on the atmospheric side, and the other on the gas side. The latter is working with a relatively small pressure difference of 0.5 to 1.0 bar between the seal oil pressure and the gas pressure, in order to keep the oil leakage towards the gas side small.
The leakage of sealing oil towards the gas side is acceptable in most cases, since the oil can easily be recovered after degasing. If required this oil leakage can be further reduced by providing the inner side of the gas side ring with spiral wind-back grooves. This special feature combines the advantages of floating ring seals (no wear, long life) with those of mechanical seals (minimum oil leakage to the gas side).



Mechanical seal

Where extremely low seal oil flow towards the gas side is required, a single or double mechanical seal may be installed. These seals can be used without reservation up to 90 m/s sliding velocity. The mechanical seal can also serve as stand-still seal.

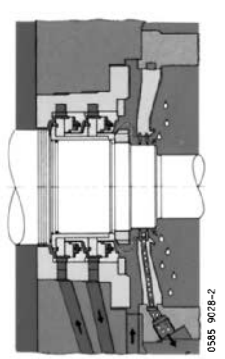
Although mechanical seals should primarily be used for applications where the ring can be periodically checked for wear, they have also been installed for services where minimum intervals between overhauls are well in excess of 10 000 operating hours.



The mechanical seal consists of a stationary carbon face mounted in a sealing ring support. It is moderatly pressed against the face of a shaft collar by several springs as well as by the sealing oil pressure, which is about 1 to 3 bar higher than the gas pressure. Symmetrically placed double mechanical seals are available. At standstill of the compressor and the seal oil pump, the negligible quantity of sealing oil required is supplied from the overhead tank to which the reference gas pressure is connected.

Gastight dry gas seal

For flammable, toxic and pure gases up to temperatures of 200 °C. One single seal is capable of sealing a pressure differential of 100 bar. For higher levels the number of seals is increased accordingly. Process gas tapped from the compressor discharge is fed to the seal through a fine (2 μ) filter. The small leakage is led to the stack. A fan ensures that an ample air flow dilutes at all times any accumulation of gas/oil vapor behind the seal to a safe level, well below an explosive mixture. An additional seal serving as standby can be installed (tandem arrangement). Installation of a labyrinth seal with inert gas injection between two dry gas seals is possible.



R Reference pressure (connected to the seal at the compressor discharge side)
O Sealant overflow for cooling of seal

Atmospheric or air side
Process gas, also sealing or buffer gas
Sealant liquid, seal oil
Sealing gas and sealant liquid mixture: seal gas to flare or back to suction via separator and/or active carbon filter; contaminated sealant to separator or collector tank to waste or to oil reclamation unit
Gas/oil vapor mixture diluted with air

FIG. C-182 Shaft seal arrangements. (Source: Sulzer-Burckhardt.)



FIG. C-183 Inlet guide vane assembly. (Source: Sulzer-Burckhardt.)

Emergency axial thrust ring. Compressors without their own thrust bearing can be fitted with an emergency axial thrust ring so that if the coupling between compressor and driver fails, the rotor remains in position as it slows down and does not rub on the casing. A relatively large clearance (approx. 1.0 mm) is provided so that the thrust ring does not rub during normal operation.

Oxygen compressors are always provided with an axial thrust ring.

Shaft seals. See Fig. C-182.

Inlet guide vanes. Figure C-183 illustrates an inlet guide vane assembly. On centrifugal compressors running at constant speed and mainly full load, a suction throttle valve is the most appropriate way to reduce the starting torque and for part-load operation. In cases where part-load occurs frequently and power is highly evaluated, inlet guide vanes achieve higher part-load efficiencies and a somewhat larger stable operating range. Inlet guide vanes before the first stage can be accommodated as a standard option of a horizontally split or the recycle stage of a vertically split compressor. They are located in the inlet channel and are adjusted through linkages by a ring that in turn can be operated by a hand wheel or connected to a pneumatic or electric actuator allowing for automatic volume or pressure control or alternatively remote setting. The vanes are provided with shafts pivoting in self-lubricated bushings. These are completely maintenance-free and due to the absence of lubricant, there is no contamination of the process gas. See Fig. C-184.

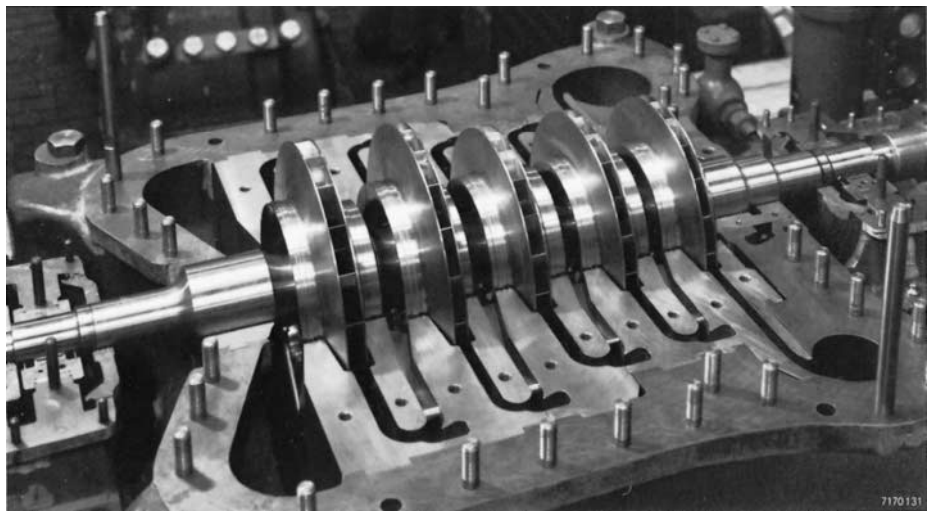


FIG. C-184 Centrifugal compressor with injection and drainage system for the compression of dirty gases. (Source: Sulzer-Burckhardt.)

Fluid injection devices. When compressing dirty gases or fluids that can cause crystal formation or polymerization, it is possible that some of these impurities might settle on the inside of the compressor channels and clog the internal passages. Injection devices have been developed for cleaning the insides of compressors, either periodically or continuously, so as to maintain the original performance.

When necessary, injection nozzles are located in the flow channels and the washing fluid is injected as close as possible to the deposits. In centrifugal machines injection is effected in all stages. Nozzles may also be provided to wash the very narrow leakage paths at the rotor seals.

The amount of fluid injected is controlled with dosemeters.

To prevent corrosion of those parts that come in contact with the fluid, conditions are controlled to avoid high temperatures, high water content due to evaporation of the water, or saturation of the process gas.

By using specially adapted materials for the internal components, gases can also be compressed in fully wet condition. Means are provided in the compressor casing for the drainage of excess fluid and sludge.

Compressor materials. Typical metallurgical selections for compressor materials are listed in Fig. C-185.

Application example: Retrofit design modification case history: Mopico™ compressor for gas pipeline stations

Updating requirements of existing installations. There are approximately 6400 gas compressor units of all types in the United States, with a total rating of 10 million kW (13 million hp) installed in over 1050 stations on the U.S. gas transmission system. These mainline stations are spaced mainly 50–70 miles apart and include 2 to 30 units. Most of these stations were built more than 30 years ago.

Such gas pipeline right-of-ways often consist of three or four pipes, 24 to 42 inches in diameter and were originally rated for 60 to 70 bar maximum operating pressure, a value that has been reduced to 50 to 60 bar due to the age of the installations.

Element	Material designation	DIN No.	ASTM or AISI equivalent
Casings	GG-20	0.6020	A 278, Class 30
	GGG-40	0.7040	A 395
	GGG-Ni 22	0.7670	A 439, Type D-2C
	GS-C25	1.0619	A 216, Grade WCB
	GS-17 CrMo 5 5	1.7357	A 217, Grade WC6
	GS-17 CrMo 9 10	1.7377	A 757, Grade D1 Q1
	G-X 5 CrNi 13 4	1.4313	A 487, Class CA6NM
	G-X 5 CrNi 17 4	–	Cast 17-4
	25 CrMo 4	1.7218	A 336, Class F12
	StE 355	1.0562	A 537, Class 1
Diaphragms	HI	1.0345	A 515, Grade 55
	GG-20	0.6020	A 278, Class 30
	GGG-40	0.7040	A 536 or A 395
	GGG-Ni 22	0.7670	A 439, Type D-2C
	GS-C25	1.0619	A 216, Grade WCB
	G-X 5 CrNi 13 4	1.4313	A 487, Grade CA 6NM
	G-X 5 CrNi 17 4	–	Cast 17-4
	X 5 CrNi 18 9	1.4301	A 276, Type 304
	X 20 Cr 13	1.4021	A 276, Type 420
	X 20 CrNi 17 2	1.4057	A 276, Type 431
Shafts	28 NiCrMoV 8 5	1.6932	A 470
	20 CrMoNiV 4 7	1.6979	A 470
	25 CrMo 4	1.7218	AISI 4130
	34 CrNiMo 6	1.6582	AISI 4340
	X 20 CrMoV 12 1	1.4922	A 565, Grade 616 HT
	X 20 CrNi 17 2	1.4057	AISI 431
	NiCu30Al	2.4375	Monel K-500
Impellers	25 CrMo 4	1.7218	AISI 4130
	23 CrNiMo 7 4 7	1.6749	A 471
	20 CrMoNiV 4 7	1.6979	A 471, Class 10
	X 2 CrNi 13 4	–	A 182, Grade F3NM
	X 5 CrNi 13 4	1.4313	A 182, Grade F6NM
	X 6 CrNiMoCu 14 5	–	A 705, S45000 (14-5PH)
	X 20 CrMoV 12 1	1.4922	A 565, Grade 616 HT
	X 20 CrNi 17 2	1.4057	AISI 431
	X 2 CrNiMoN 22 5	1.4462	A 182, Grade F51
	NiCu30Al	2.4375	Monel K-500
	TiAl6V4	3.7165	B 381, Grade F-5
Labyrinths	X 6 CrMo 17 1	1.4113	AISI 434
	X 6 CrNiTi 18 10	1.4541	AISI 321
	NiCu30Fe	2.4360	Monel 400
	Titanium	3.7055	Titanium

FIG. C-185 Typical compressor material selection. (Source: Sulzer-Burckhardt.)

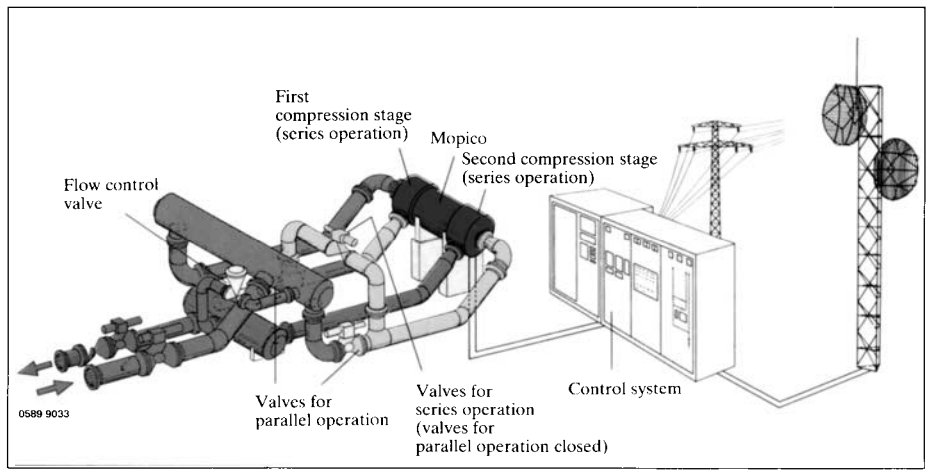


FIG. C-186 Schematic layout of a compressor station. (Source: Sulzer-Burckhardt.)

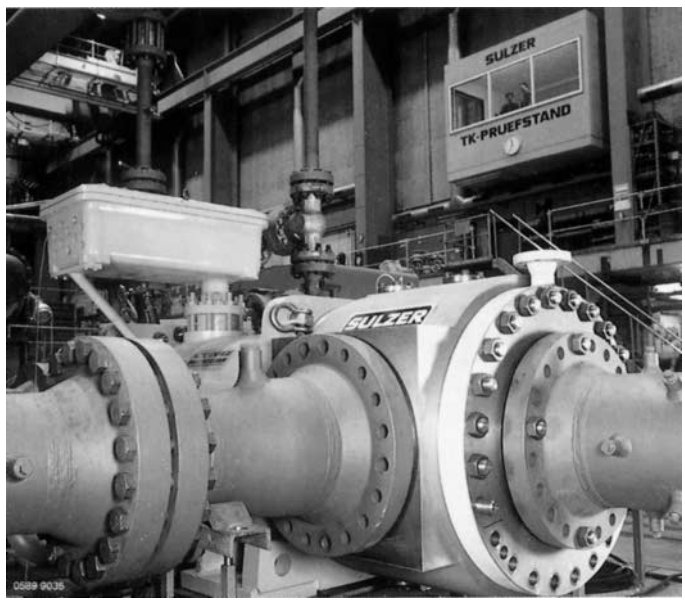


FIG. C-187 Mopico compressor on a test bed. (Source: Sulzer-Burckhardt.)

These antiquated systems are very limited in their delivery capacity compared to modern systems. As an example, a 55-mile pipeline section between stations, having one 30-, two 36-, and one 42-in conduits operating at a maximum line pressure of 55 bar can transport about 100 million Nm^3 gas per day. Some 15 to 20 individual integral gas engines, drive rating 33,000 to 37,000 kW are required for this purpose.

Such installations compare poorly with modern systems: a typical Russian gas pipeline built in the 1980s can transmit the same 100 million Nm^3 per day through 55 mi of a single 56-in conduit at 74.5 bar maximum line pressure using two gas turbines producing the same total output.

The Russian compressor stations are standardized in design and rating and are located approximately every 65 mi and include over 75,000 kW in gas turbine power. Because the Russian system was developed 30 years after the American, the gas

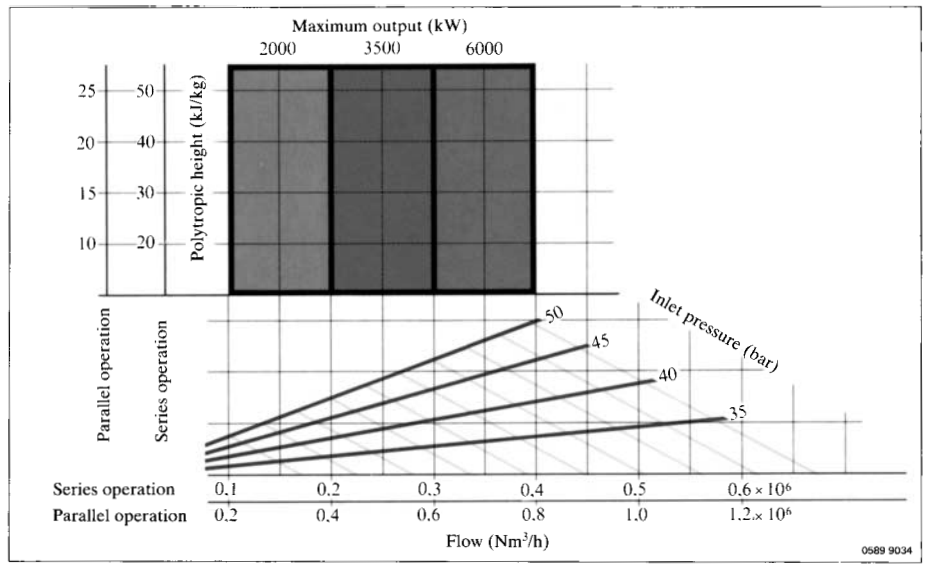


FIG. C-188 Operating ranges of the Mopico compressor. (Source: Sulzer-Burckhardt.)

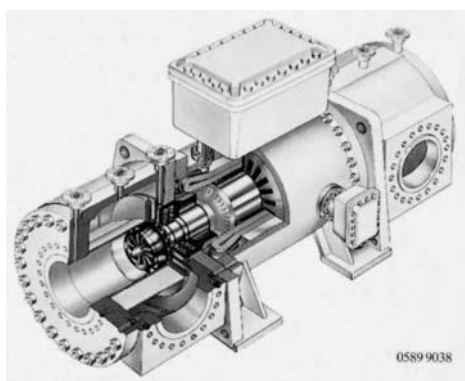


FIG. C-189 Cutaway section through the new Mopico gas pipeline compressor. Motor and compressor are fitted into a hermetically sealed casing. (Source: Sulzer-Burckhardt.)

turbine technology of the 1970s was available to them, as well as the large diameter, high pressure pipe produced in Japan and Europe.

The European pipeline network that was built around 1970 is on a par with other systems built in the same period, i.e., the Argentinian, Australian, and Canadian systems. However, during the past few years, the Canadian gas pipeline companies have carried out tests with new gas turbine units and combined cycle systems. Moreover, the first high-power motor compressors using variable-frequency drives (VFDs) have been installed. These 6-MW and 18-MW systems use German and Swedish synchronous motors and the load commutated inverter (LCI) drive technology.

In contrast, the gas pipeline networks in the United States have had virtually no important new pipeline technology applied to them. The new Mopico compressor, with its high-speed induction motor, magnetic bearings, and variable-frequency drive, is useful for the conditions in older installations.

See Fig. C-186 for a typical compressor station layout, Fig. C-187 shows one of these compressors on a test bed, and Fig. C-188 shows the operating ranges of the "Mopics."



FIG. C-190 Rotor shaft with radial impellers. (Source: Sulzer-Burckhardt.)

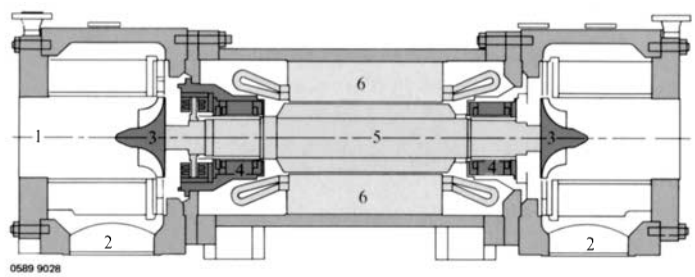


FIG. C-191 Section through the simply constructed Mopico compressor. (Source: Sulzer-Burckhardt.)

Design. The Mopico gas pipeline compressor (see Figs. C-189 and C-190) features a high-speed, two-pole, squirrel-cage induction motor. Motor and compressor are housed in a hermetically sealed, vertically split, forged steel casing (Figs. C-189 and C-191). The center section contains the motor and bearings, and each of the end casing sections houses a compressor wheel, a fixed vane diffusor, and the inlet and discharge flanges.

Mopico compressors can be operated in series or parallel. Magnetic radial bearings and a double-acting magnetic bearing maintain the runner in position. Figure C-190 shows the rotor shaft with radial impellers. The motor is cooled by gas metered from the high-pressure plenum of one of the compressor housings. Hence the Mopico runs completely oil-free.

The speed and thus the discharge rate of the Mopico unit is controlled by a thyristorized, variable-frequency drive. This drive uses thyristors that can be switched out. These enable pulse-free runup without current peaks and an operating speed range of 70 to about 105 percent.

Design criteria. The following conditions can be complied with through the new combination of elements:

- Low installation, maintenance, and energy consumption costs
- Broad operating range at high economic performance
- Compatibility with existing compressors
- Unattended remote control

- Emissionless and oil-free
- Can be installed outdoors

Based on cost per installed hp, the cost of the Mopico compressor is only about two-thirds that of a gas turbine unit and less than half that of a low-speed reciprocating compressor. It is some 10 percent less than a "conventional" centrifugal with dry seals, magnetic bearings, and a direct-drive, high-speed induction motor with variable-frequency drive.

On the other hand, a conventional centrifugal with motor-gear drive and IGV is less expensive. This system is, however, unacceptable for pipeline application because of poor efficiency at low pressure ratio conditions.

The overall energy consumption costs of a Mopico system using energy produced in a base load power station is considerably lower than the cost of either a gas engine or a gas turbine burning natural gas.

Gas pipelines impose the most stringent operating requirements. A typical main line station has to accommodate flow differentials of 50 percent and more between winter and summer. In order to handle the wide range of part-load conditions most efficiently, a main line station should include multiple individual units, each with a broad operating range at high efficiency. Both prime mover and compressor have to be taken into account when evaluating part-load efficiency.

Most of the competitive systems offer good design point efficiencies, but show rapid efficiency deterioration below 70 percent load. With Mopico, however, the motor-VFD system operates within 3–5 percent of its design efficiency over the full operating range of the pipeline. Motor speed never drops below 70 percent, since the operating mode of the system changes from series to parallel at that point. A mainline station with six Mopico units would typically operate six months of the year with three units in the parallel mode. For the rest of the year, the station would operate with four to six units in the series mode.

Compatibility. There is no problem associated in paralleling Mopico units with existing recips or centrifugal compressors. The speed of the Mopico is adjusted to attain the desired level of loading.

Unattended remote control. The system is designed to be operated by remote signals much the same as motor pumps on oil pipelines.

Low maintenance cost. A Mopico compressor has no wearing parts and thus requires practically no regular maintenance. Planned maintenance is limited to the replacement of the water cooling pump's mechanical seal every few years, the periodic cleaning of the filters for the control room air-conditioning system and the normal verification and adjustment work on the electronic control equipment of the drive system and of the magnetic bearings.

Oil-free, no emissions, intrinsically safe. The Mopico has no shaft seals, because it runs on magnetic bearings. Therefore there is no oil requirement in the system. Moreover, since there is no combustion, there are no emissions. Intrinsic safety of the system arises from the hermetic sealing and the fact that the motor is fully pressurized with cooling gas (no air).

Installation. The Mopico compressor can be installed out- or indoors. The VFD system, magnetic bearing controls, other unit controls and switchgears are designed for installation in a weather-protected building outside of the hazardous area.

Design features

Motor. The electric motor manufacturer selected an asynchronous motor for the Mopico system. The advantages of such high-speed, squirrel-cage motors are:

- Good efficiency
- Low maintenance requirements
- Ability to comply with the safety regulations for hazardous areas
- Axisymmetric rotor
- Simple construction

Additional specific design objectives for this particular application were:

- Ability to function in a gas-filled environment
- Compatibility with the variable-frequency drive
- Compatibility with the magnetic bearings

The main terminal box is located on the top of the motor housing. Two smaller terminal boxes on both sides are used for controlling and feeding the magnetic bearings. Both terminal box types meet the requirements of the various safety regulations for hazardous areas (CSA, NEC, EuroNorm).

Because the motor is cooled by gas, the stator windings are built to Class H standards, with mica and glass-based insulation materials that withstand temperatures to 200°C. The impregnation technique uses a silicone resin to ensure that the stator can completely withstand mechanical stresses and remain impervious to dampness and corrosive environments. Special shields in nonmagnetic materials are then adjusted inside the housing to avoid eddy-current losses in the steel.

Compressor. The compressor module consists of three main parts: the outer, pressure-bearing discharge casing, the inlet insert and the radial compressor impeller. The outer casing is machined from a single steel forging with an integral discharge flange. The symmetry of this and the motor casing allows the discharge flange to be arranged at any angle. There are no lining-up problems, since the solid casing makes the unit impervious to forces associated with the external pipeline. The insert is a simple welded structure, which combines the functions of axial inlet to the impeller, inlet labyrinth carrier, and diffusor carrier. Attachment of the radial impeller to the high-speed motor shaft end is by means of a central tie bolt; torque is transmitted by means of two drive pins.

The impellers are of the shrouded type and belong to the family of modern impellers designed in recent years using the latest numerical design and testing techniques. A wide operating range at high efficiency is a feature of such impellers.

The diffusor was developed especially for the Mopico unit. It has a rather short radial section containing fixed vanes in a tandem arrangement. At the exit to the vanes, the now purely radial flow is dumped into an annular space. In comparison to more conventional diffusors, the Mopico type exhibits a flat loss characteristic.

Drive. The frequency control equipment is located in a separate building, outside the area classified as hazardous. Also housed inside the building are the motor starter-breaker, the magnetic bearings control system (including an auxiliary power supply), the Mopico unit controls, including valve sequencing, auto start/stop, surge control, real-time performance monitoring, and the various system monitoring, alarm, and shutdown devices.

Testing of the Mopico system. Already at the beginning of the design phase of the motor, tests were carried out with a small test rotor having the same diameter as the actual one, but reduced in length. The purposes of these tests were to study the mechanical stresses in the rotor and to evaluate the windage losses. This rotor was mechanically driven up to 13,000 rpm, without any permanent deformation occurring.

TABLE C-13 Test Results, Design, and Maximum Values

	Test	Design	Maximum
Speed (rpm)	9,450	9,850	9,850
Shaft power (MW)	5.63	5.74	8.5
Line voltage (kV)	5.74	5.94	5.94
Current (A)	688	677	979
Power factor	0.87	0.87	0.88
Motor efficiency (%)	94.8	95.1	95.6
Losses (kW)	309	293	387
Cooling gas	N_2/He	Methane	Methane
Inlet temperature (°C)	57	33	33
Outlet temperature (°C)	103	60	69
Heat capacity (J/kg°C)	1,207	2,500	2,500
Flow (% of compressor flow)	2.6	2.0	1.3
Motor temperature rise			
Stator winding mean T (°C)	71	57	122
Stator winding mean T (°C)	128	90	155
Stator winding max. T (°C)	140	100	175
Rotor cage mean T (°C)	120	75	117

During the manufacturing phase, specific tests were performed on some components: motor casing, compressor casing, and pass-throughs were tested under pressure up to 150 bar by INIEX, the Belgian control laboratory, or by equivalent laboratories in the U.S., and accepted. Insulation and winding components were tested in a natural gas environment.

The Mopico compressor on the test bed is shown in Fig. C-187. Dummy impellers, having the same mechanical characteristics as the real ones, were used for the no-load conditions. It was therefore possible to tune the magnetic bearings in the full-speed range, to measure the no-load losses (mechanical and iron losses), and to have better knowledge of the electrical parameters necessary to compute the performances of the motor.

Finally, the whole prototype was tested on-load. Several test runs were performed in order to obtain thermal stabilization of the whole circuit and particularly of the motor. These test runs were made under various conditions of speed, pressure, load, nature of gas, level of cooling flow in the motor, and operation mode of the control loop.

A test performed at 9500 rpm and 5.63 MW shaft power in series mode with a mixture of nitrogen and helium was very close to the design nominal conditions (9850 rpm, 5.74 MW). Table C-13 summarizes the test results. Comparing the first two columns shows the influence of the nature of the gas and of an efficient cooling on the motor temperature rise. With 23 percent less relative cooling flow, the temperature rise is reduced by 20 percent. By virtue of the large safety margin in temperature, it could be possible either to increase the output power or to decrease the cooling flow.

The last column shows a theoretically available shaft power of 8.5 MW without excessive motor temperature rise and with a constant absolute cooling flow.

Interaction of magnetic bearings and rotor. Magnetic bearings have, besides the well-known advantages of no wear, no lubrication, and low power consumption, also the reputation of being able to solve any rotordynamic problem. This is not true. On the contrary, they can cause a variety of new problems.

The Mopico compressor has, at present, the heaviest rotor running at a speed of 10,000 rpm on magnetic bearings. The stiffness and damping coefficients of

magnetic bearings depend on the vibration frequency, whereas for oil bearings they depend on the rotating speed. Compared to oil bearings, magnetic bearings have a lower stiffness. In the frequency range of 50 to 200 Hz, the stiffness of the magnetic bearings is only about one sixth of the stiffness that an oil bearing for Mopico would have at a speed of 10,000 rpm. In order to prevent large static deflections due to the low stiffness, the controller of the magnetic bearing has an integration term.

Phase lead cells in the controller provide the damping of the magnetic bearing—however, only in a limited frequency range. At very low frequencies (below about 30 Hz) and at very high frequencies (above 1500 Hz), the damping is negative. In the frequency range of 50 to 200 Hz, the damping coefficient of the magnetic bearing is of the same order of magnitude as that of an oil bearing.

The theoretical model of the rotor bearing system is well proven by measured closed loop transfer functions (relation of the displacement at the sensor and a magnetic excitation force at the bearing).

The rotor can be safely run up to the design maximum speed of 10,000 rpm. The vibration level at this speed is not more than about 50 percent of the limit due to amplifier saturation. If the cold rotor is run up slowly to this speed (within 10 to 15 min), the level is even lower.

Isotherm turbocompressors

Turbocompressors with the lowest power consumption. The word *isotherm* describes the principal feature of these machines; the flow medium is cooled intensively during the compression process in order to come as close as possible to ideal isothermal compression, giving maximum efficiency and therefore minimum power requirement.

Isotherm compressors are particularly suited to oil-free compression of air, oxygen, or nitrogen to discharge pressure up to 13 bar. For higher pressure ratios a booster may be added to the compressor train. The isotherm compressors are widely used in air separation plants, fertilizer plants, iron and steelworks, and for compressed-air supplies to mines. See Figs. C-192 and C-193 for illustration of some typical applications. The design has been steadily improved since they were first introduced in 1913. An important factor for their great success is their low specific power requirement, a result of flow path and intensive intercooling. Compact, standardized construction and high availability are required of them. More than 1000 isotherm compressors are in service throughout the world, many with up to 200,000 h to their credit and a time between overhauls of three to six years.

Operating range. See Fig. C-193.

Design features. Design features are incorporated to suit an end user's application. Typically they include:

1. Low power consumption
2. Resistance against corrosion
3. High rotor stability = low vibration level
4. Low noise level
5. Shaft-string configuration
6. Ease of installation
7. Simple maintenance programs
8. Minimized space requirement
9. High reliability

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